

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

401
12
December 2022

Course Code: **MAT101R01**

Course: **ENGINEERING MATHEMATICS-I**

QP No. : **U183A**

Duration: **3 hours**

Max. Marks: **100**

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. Test whether $\operatorname{sech}^{-1} x = \pm \log \left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1} \right)$ provided $x \neq 0$ or not.
2. If $4 \cosh x - 3 \sinh x = R \cosh(x - y)$, then obtain the value of R and y .
3. Obtain the principal logarithm of $1 + i$.
4. For what values of x , the matrix $\begin{pmatrix} 10 & x & 0 \\ 1 & 5 & 2 \\ 3 & 1 & 1 \end{pmatrix}$ has the rank 2 and 3?
5. If 1, 2, 3 are eigen values of A , then obtain the sum and product of eigen values of the inverse of transpose of $2A$.

18. (a) Derive the value of $\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx$ in terms of beta function

and hence evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \cos^6 x dx$ (8)

(b) Find the area of the region D bounded by the parabolas $y = x^2$ and $x = y^2$. (7)

PART- C

Answer the following

1 x 20 = 20 Marks

19. (a) Convert the quadratic form $x^2 + y^2 + z^2 + 2yz$ into canonical form by using orthogonal reduction and hence classify its nature. (10)

(b) Test whether the $\log \left[\frac{(1+i)(2+i)}{3+i} \right]$ is real or not. (5)

(c) Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ by using triple integral. (5)

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(b) Obtain the inverse of $\begin{pmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix}$ by using elementary transformations. (7)

(OR)

14. (a) Verify that the eigen vectors of $\begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$ are orthogonal in pairs. (8)

(b) Obtain the singular value decomposition of the matrix $\begin{pmatrix} 4 & 4 \\ -3 & 3 \end{pmatrix}$ (7)

15. (a) Diminish the roots of $x^4 - 4x^3 - 7x^2 + 22x + 24 = 0$ by 1 and Hence solve it. (8)

(b) Find the sum to ∞ of the series $1 + \frac{1}{5} + \frac{1.4}{5.10} + \frac{1.4.7}{5.10.15} + \dots$ (7)

(OR)

16. (a) Solve $6x^5 + x^4 - 43x^3 - 43x^2 + x + 6 = 0$. (8)

(b) Find the sum to ∞ of the series $\frac{2.3}{3!} + \frac{3.5}{4!} + \frac{4.7}{5!} + \dots$ (7)

17. (a) Evaluate $\int_0^a \int_y^a \frac{x \, dx \, dy}{x^2 + y^2}$ by changing order of integration. (8)

(b) Using gamma function, find $\int_0^1 (x \log x)^3 dx$ (7)

(OR)

6. Justify your comment on the following statement. If a and b are real numbers such that $a > b$, then the matrix $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ has the eigen values $a + b$ and $a - b$ with opposite signs.
7. Solve $x^3 - 13x^2 + 15x + 189 = 0$ whose two roots differ by 2.
8. Find the coefficient of x^n in the expansion of $(1 + x + x^2 + \dots)^2$.
9. Obtain the value of I by using double integral where I is area of the square with vertices $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$.
10. Express the integral $\int_0^\infty \left[\log \left(\frac{1}{x} \right) \right]^{n-1} dx$ in terms of gamma function.

PART -B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Prove that $\frac{1 + \cos 7\theta}{1 + \cos \theta} = (x^3 - x^2 - 2x + 1)^2$ provided $x = 2 \cos \theta$. (8)
- (b) Expand $\sin^3 \theta \cos^5 \theta$ in a series of sines of multiples of θ . (7)

(OR)

12. (a) Obtain the real and imaginary parts of $\tan^{-1}(x + iy)$ (8)
- (b) Derive the series expansion of $\tan \theta$ in ascending powers of θ up to θ^7 . (7)
13. (a) Test the consistency of the system of equations $2x + 4y + 2z = 1$, $x + y + z = 2$, $x - y + z = 1$ and hence solve if it is consistent. (8)